

Neural network-based analysis of charge exchange spectra in KSTAR

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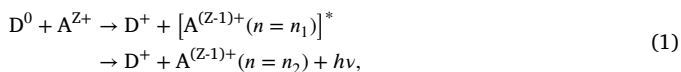
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ABSTRACT

Charge exchange spectroscopy (CES) is widely used to measure ion temperature, density, and rotation velocity profiles in various fusion devices. Conventional CES diagnostics typically rely on beam modulation techniques to separate active charge exchange components from raw spectral signals. However, in KSTAR, beam modulation is performed using the main heating beam rather than a dedicated diagnostic beam, which may potentially cause plasma disturbances. To circumvent the need for this perturbative technique during analysis, we have developed a neural network-based model (NN-CES) that infers ion temperature and toroidal rotation velocity directly from raw measured spectra. The model employs a physics-constrained architecture that optimizes prediction through combined parameter and spectral shape consistency losses. It is trained on an extensive dataset collected under representative plasma conditions in KSTAR, including low-confinement mode (L-mode), high-confinement mode (H-mode), locked mode, and resonant magnetic perturbation (RMP) application. In this work, we demonstrate that NN-CES predictions are in good agreement with results from conventional beam modulation analysis, enabling modulation-free diagnostics with the potential for real-time plasma control.

1. Introduction

In magnetically confined fusion plasma experiments, charge exchange spectroscopy (CES) has been used for decades to diagnose ion temperature, density, and rotation velocity profiles by analyzing charge exchange (CX) interactions between neutral beam atoms and plasma or impurity ions [1–8]. The CX process follows the reaction:



where D is an injected neutral beam atom, typically deuterium or hydrogen; A is a plasma or impurity ion with an elementary charge Z; the asterisk denotes an excited state of the ion after CX interaction; and $h\nu$ is the characteristic photon emitted during the spontaneous decay of the excited ion. Here, n_1 and n_2 are the principal quantum numbers of the excited and lower energy states, respectively, and n refers to the principal quantum number. During the CX process, the neutral beam atom donates an electron to an ion species, potentially leaving it in an excited state. The excited ion then undergoes spontaneous emission, releasing characteristic radiation in the form of a photon. This CX emission is collected by a spectrometer for analysis.

The CX measurements are spatially localized to the intersection of the diagnostic line of sight (LOS) and the trajectory of the neutral beam injection (NBI). However, since the measured signal is line-integrated along the LOS, it contains not only the active CX signal but also background components originating from other processes. In particular, thermal neutral particles – of the same species as the injected beam neutrals but present in the plasma edge – undergo the CX process with plasma or impurity ions, producing additional spectral lines that overlap with the active CX signal. Moreover, electron impact excitation of these ions contributes further background emissions. To accurately extract the active CX signal from the measurements, these background components should be carefully subtracted.

A widely used method for background subtraction in many fusion devices is beam modulation, in which the neutral beam is periodically turned off to directly measure the background emission. To estimate the background during the beam-on phase, a spectrum is typically generated by interpolating between the measurements from the preceding and succeeding beam-off intervals. This technique provides a reliable background estimate in steady-state plasmas, where conditions remain stable. However, in scenarios with rapidly evolving plasma dynamics

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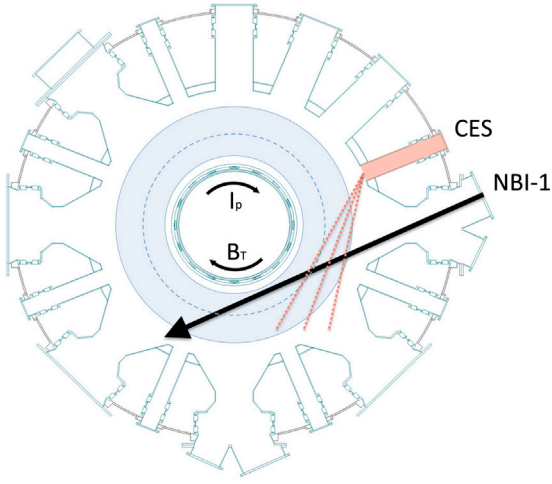


Fig. 1. Schematic top view of the KSTAR CES diagnostic system, showing the line of sight (red dashed) and the injection line of NBI-1 (black solid arrow). In KSTAR, both the plasma current (I_p) and toroidal magnetic field (B_T) are usually in the clockwise direction.

or low signal-to-noise ratios, this estimate may not fully represent the actual background present during the beam-on phase, leading to potential uncertainties in the subtracted active signal. Additionally, frequent modulation of the neutral beam can disturb plasma conditions, affecting the accuracy of CES measurements.

Based on an extensive CES dataset accumulated from beam modulation-based background subtraction analysis in KSTAR, we propose a neural network-based analysis of charge exchange spectra (NN-CES). While previous studies have applied neural networks to CES data using curve-fitting approaches for background estimation [9–11], our approach, in contrast, trains a model to directly predict the ion temperature and toroidal rotation velocity from the raw spectra, as obtained in a conventional beam modulation analysis. This method trains a neural network model that learns the relationship between the raw measured spectra and the Doppler shift and broadening parameters, which are used to derive toroidal rotation velocity and ion temperature, respectively. Once trained, the model can directly infer these parameters from the raw spectra, eliminating the need for beam modulation. In this work, we present the development of the NN-CES model, detailing its training procedure. The performance of the trained model is evaluated by comparing predictions with beam modulation results, demonstrating its potential as an alternative approach for background subtraction.

2. The KSTAR CES diagnostic

2.1. System overview

Fig. 1 shows a schematic view of the KSTAR CES diagnostic system, which is aligned with the trajectory of the NBI-1. A spectrometer with a pitch-controlled double-slit fiber bundle is utilized in the KSTAR CES system, effectively doubling the number of available fibers by dividing the charge-coupled-device (CCD) image into two spectral regions [12]. However, since the spectral images from the two columns of fibers overlap, a band-pass filter is applied to each channel to prevent interference. Consequently, the system provides 32 spatially resolved channels at the midplane, covering the radial region from the magnetic axis to the plasma boundary on the low-field side. Its spatial resolution is 20 mm in the plasma core and 5 mm in the edge region.

In KSTAR, the wall components are primarily composed of carbon, while NBI-1 injects a deuterium beam. When the beam is injected, CX

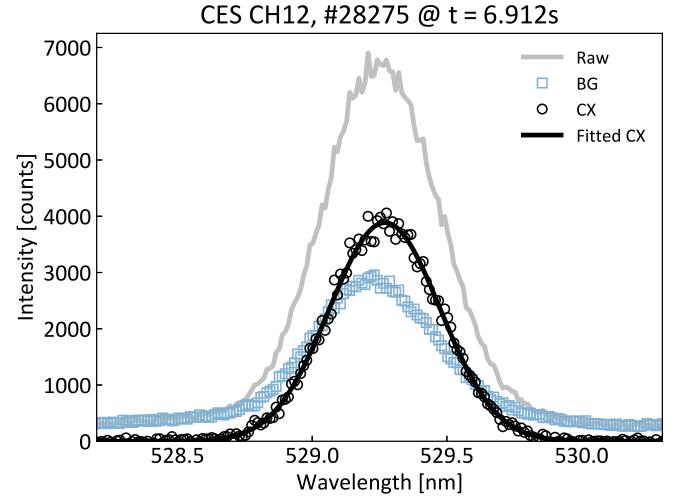
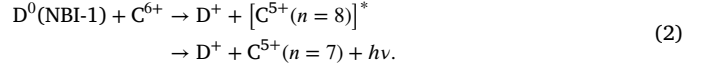


Fig. 2. Analysis of CX spectrum using the conventional beam modulation technique. The data is from KSTAR #28275 at $t = 6.912$ s, measured at CH12 ($R = 2.147$ m). The raw spectrum during the beam-on phase (gray line), the background during the beam-off phase (blue squares), the resulting active CX spectrum after subtraction (black circles), and the final Gaussian fit to the active signal (black line) are shown.

interactions between carbon ions and deuterium beam neutrals occur actively in plasmas. The CX process is given by:



The KSTAR CES system detects C_{VI} ($n : 8 \rightarrow 7$) emission at the rest wavelength of $\lambda_0 = 529.05$ nm for spectroscopic analysis. It thus provides the temperature and flow velocity profiles of C^{6+} ions [13], as well as their density profile [14].

2.2. Background subtraction

The KSTAR CES system routinely employs the neutral beam modulation technique to subtract background components from the measured signals. Typically, the neutral beam is turned off periodically at 1 Hz with a 10 ms beam-off interval. During the beam-on phase, the measured spectrum contains both active CX emission and background emissions, while the beam-off phase captures only the background. Subtracting the beam-off background spectrum from the beam-on measurement yields the active CX emission, as illustrated in **Fig. 2**.

The extracted active CX spectrum, denoted as $I_{CX}(\lambda)$ at the wavelength λ , is typically described by a Gaussian function:

$$I_{CX}(\lambda) = A(\lambda_0) \exp \left[-\frac{(\lambda - \lambda_0 - \Delta\lambda)^2}{2W_D^2} \right], \quad (3)$$

where $A(\lambda_0)$ is the spectral amplitude at λ_0 corresponding to the CX transition, $\Delta\lambda$ is the Doppler shift from λ_0 , and W_D is the Doppler broadening of the spectral line. The parameters (A , $\Delta\lambda$, and W_D) are obtained by applying nonlinear least-squares fitting techniques to the background-subtracted CX spectra.

The Gaussian fitting parameters extracted from the CX spectra – Doppler broadening and Doppler shift – are directly related to ion temperature and flow. In particular, if the ion population exhibits a net flow along the LOS, a Doppler shift of the spectral line from its rest wavelength is observed. In the KSTAR CES system, where the LOS is nearly aligned with the toroidal direction, this shift corresponds to the toroidal rotation velocity. Meanwhile, the Doppler broadening reflects the thermal spread of ion velocities and is used to determine the ion

temperature. These relationships are expressed as:

$$\begin{aligned} v_T &= c \frac{\Delta\lambda}{\lambda_0}, \\ T_i &= m_i c^2 \left(\frac{W_D}{\lambda_0} \right)^2, \end{aligned} \quad (4)$$

where T_i is the ion temperature, v_T is the toroidal rotation velocity, c is the speed of light, and m_i is the ion mass of the measured species (CvI ions, in this case). W_D represents the pure Doppler broadening, obtained after correcting the fitted width for the fixed instrumental width of the spectrometer. In KSTAR, where the ion temperature is typically well above 200 eV, Zeeman splitting effects are estimated to remain within the overall measurement uncertainty under representative plasma conditions.

Despite its widespread usage, the beam modulation approach has inherent limitations. First, the background measured during beam-off phase may not accurately represent the actual background present during beam-on phase, particularly in rapidly evolving plasma conditions. Second, the periodic interruption of the neutral beam can disturb plasma conditions, potentially affecting the reliability of the diagnostic measurements. Third, because the background emission during the beam-on phase is estimated by interpolating between adjacent beam-off phases, this method requires waiting for the next beam-off interval, making fast-response analysis difficult.

These limitations have motivated the development of an alternative background subtraction approach that does not rely on beam modulation. Based on the extensive dataset of CES measurements obtained from beam modulation experiments in KSTAR, we develop the NN-CES model, which infers ion temperature and toroidal rotation velocity directly from raw measured spectra, thus avoiding the need for beam modulation.

3. NN-CES for KSTAR

3.1. Neural network model

Fig. 3 illustrates the proposed NN-CES model, which employs a physics-constrained neural network to learn the mapping between the measured spectra – comprising both active CX and background emission – and the corresponding Doppler shift and Doppler broadening of the CX spectral line. Once trained, the NN-CES model can directly predict these parameters from raw spectra, enabling rapid inference of the ion temperature and toroidal rotation velocity without the need for beam modulation.

The neural network architecture consists of a multi-layer perceptron (MLP) with fully connected layers. The input layer consists of 256 neurons corresponding to the raw spectral data points, followed by two hidden layers with 256 neurons each, and an output layer that predicts two parameters of the CX spectral line: the Doppler shift and Doppler broadening. Each layer employs the Rectified Linear Unit (ReLU) activation function [15]. Prior to training, the input spectra are normalized on a frame-by-frame basis by dividing each spectrum by its maximum value, which leads to more stable predictions.

3.2. Loss functions

During training, the neural network model's optimization is guided by minimizing a total loss function, which is a weighted sum of three distinct components: a loss for the Doppler shift, a loss for the Doppler broadening, and a physics-informed loss for the spectral shape. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = \lambda_{\Delta\lambda} \mathcal{L}_{\Delta\lambda} + \lambda_{W_D} \mathcal{L}_{W_D} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}}, \quad (5)$$

where $\mathcal{L}_{\Delta\lambda}$, $\mathcal{L}_{W_D}$, and $\mathcal{L}_{\text{phys}}$ are the individual loss components, and $\lambda_{\Delta\lambda}$, λ_{W_D} , and λ_{phys} are their respective weighting factors.

Table 1

Range of key plasma parameters in training dataset.

Parameter	Range
Plasma Current, I_p	300–700 kA
Toroidal Magnetic Field, B_T	1.6–2.9 T
Line-Averaged Electron Density, $\bar{n}_e$	$1.0\text{--}5.6 \times 10^{19} \text{ m}^{-3}$
NBI-1 Injected Power, $P_{\text{NBI-1}}$	1.0–4.0 MW

The individual parameter losses, $\mathcal{L}_{\Delta\lambda}$ and $\mathcal{L}_{W_D}$, quantify the mean squared error between the predicted parameters and the target values derived from the conventional analysis:

$$\begin{aligned} \mathcal{L}_{\Delta\lambda} &= \frac{1}{N} \sum_{i=1}^N (\Delta\lambda_i - \hat{\Delta\lambda}_i)^2, \\ \mathcal{L}_{W_D} &= \frac{1}{N} \sum_{i=1}^N (W_{D,i} - \hat{W}_{D,i})^2, \end{aligned} \quad (6)$$

where N denotes the number of training samples, $\Delta\lambda_i$ and $W_{D,i}$ are the predicted parameters, and $\hat{\Delta\lambda}_i$ and $\hat{W}_{D,i}$ are the target parameters.

The third term, the physics loss ($\mathcal{L}_{\text{phys}}$), acts as a physics-based constraint. It ensures that the Gaussian spectrum reconstructed from the predicted parameters is physically consistent with the target spectral shape. Both spectra are normalized before comparison to focus the loss on relative spectral features. It is defined as:

$$\mathcal{L}_{\text{phys}} = \frac{1}{N} \sum_{i=1}^N \left[\frac{1}{M} \sum_{j=1}^M (I_{\text{fit}}(\lambda_j; \Delta\lambda_i, W_{D,i}) - \hat{I}_{\text{fit}}(\lambda_j))^2 \right], \quad (7)$$

where M is the number of spectral data points, I_{fit} is the reconstructed spectrum from the predicted parameters, and $\hat{I}_{\text{fit}}$ is the target spectrum. The weighting factors (λ) are hyperparameters that balance the different objectives and were empirically optimized using the validation set.

3.3. Dataset and training

As the neutral beam is attenuated during its propagation through the plasma, the CX signal becomes weaker toward the plasma core compared to the edge. To account for this variation, separate neural network models are trained for each of the 32 CES channels in this work. Additionally, channel sensitivity is affected by hardware-specific factors such as optical alignment, fiber transmission efficiency, and detector response. Training individual models per channel allows NN-CES to capture these effects and learn the unique spectral characteristics of each channel.

An extensive dataset was compiled to train and evaluate these models. In total, data from 321 KSTAR discharges, amounting to 150,890 valid spectral frames after physics-based filtering, were utilized. This dataset covers a broad representation of plasma scenarios, including low-confinement mode (L-mode), high-confinement mode (H-mode), locked mode, and discharges with resonant magnetic perturbations (RMPs). To quantitatively define the scope of this data for training, the range of key plasma parameters is summarized in Table 1.

To ensure a robust evaluation, this dataset was partitioned on a shot-by-shot basis into a training set (80%), a validation set (10%), and a test set (10%). The training set was used to optimize the model's weights, while the validation set served to guide hyperparameter tuning, such as optimizing the loss weighting factors, and to monitor for overfitting. The test set, consisting entirely of discharges not included in training, was held out exclusively for the final performance assessment reported here.

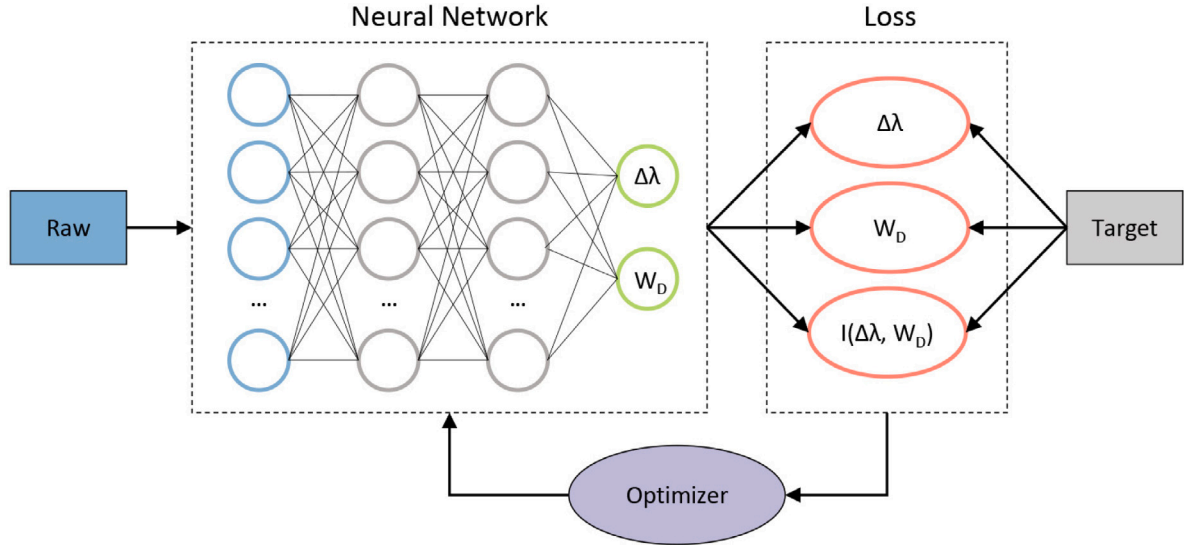


Fig. 3. Diagram of the NN-CES architecture. Raw spectral signals are processed through fully connected neural network layers to predict Doppler shift and Doppler broadening. The model is optimized by minimizing a total loss function that compares these predictions to target values derived from the conventional beam modulation analysis. This loss is composed of three weighted components: the Doppler shift loss, the Doppler broadening loss, and the physics-informed loss for the spectral shape.

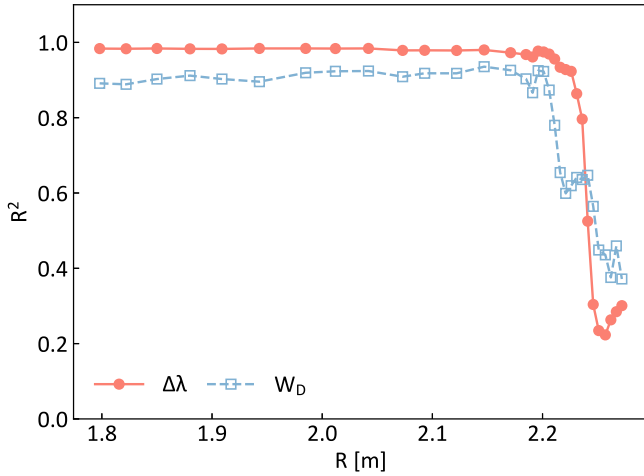


Fig. 4. Coefficient of determination (R^2) for the model's prediction of Doppler shift (red closed circles with a solid line) and Doppler broadening (blue open squares with a dashed line), plotted as a function of major radius. The evaluation was performed on the held-out test set.

3.4. Model performance evaluation

The NN-CES model's generalization performance was quantitatively assessed on the held-out test set using the coefficient of determination (R^2). Fig. 4 shows the R^2 values for the model's direct outputs – Doppler shift and Doppler broadening – as a function of major radius. The model shows strong predictive performance within the plasma interior ($R < 2.2$ m). A notable decline in performance is observed toward the plasma edge.

The reduced R^2 values observed at $R \gtrsim 2.2$ m correspond to the plasma edge region, where spectral analysis is inherently more challenging. In this region, the active CX signal is reduced due to the increased passive emission of thermal neutrals. The spectra are also more easily distorted by transient events such as ELMs and by discrepancies between beam-on and beam-off phases. These factors degrade the fitted parameters used for training and thus lower the prediction accuracy of the model in the edge channels.

3.5. Error analysis

To quantify the model's prediction uncertainty, we conducted a channel-wise error analysis on the test set. For each channel, we first calculated the residuals between the predicted and target values of $\Delta\lambda$ and W_D . To obtain robust error estimates, statistical outliers were removed from the residuals using the interquartile range (IQR) method. The standard deviation of the filtered residuals was then computed, providing a measure of the model's typical prediction uncertainty for each parameter. This statistical uncertainty is propagated to the final physical quantities and visualized as error bars in the following section.

4. NN-CES prediction results

To demonstrate the performance of the trained NN-CES model, we present a case study using KSTAR discharge #28275 from the test set. This discharge is particularly valuable as it includes a phase with applied RMPs, providing a useful benchmark for the model's capability to capture dynamic variations in plasma profiles.

Fig. 5 shows the time evolution of key plasma parameters and the corresponding NN-CES predictions at two representative radial locations: a channel near the magnetic axis and a channel near the pedestal top. As shown in Fig. 5(a) and (b), RMPs are applied from $t = 5.0$ s, leading to changes in the plasma density. At both radial positions shown in Fig. 5(c) and (d), the NN-CES model closely reproduces the overall temporal evolution of both T_i and v_T , including the distinct decrease in v_T due to the RMP application.

The full radial profiles of T_i and v_T at two time points – an H-mode phase and a later RMP phase – are compared in Fig. 6. The NN-CES predictions agree well with the target values across the entire profile from the magnetic axis to the last closed flux surface (LCFS). A reduction in both temperature and velocity is observed following the RMP application, which is accurately captured by the model.

Fig. 7 illustrates a raw spectrum together with its decomposition into active CX and background components. The NN-CES prediction, reconstructed from the inferred Doppler parameters, is shown alongside the beam modulation result. Both methods reproduce the active CX spectrum with good agreement, while NN-CES simultaneously captures the background emission present in the raw spectrum, demonstrating that NN-CES reconstructs the active spectrum while also reproducing

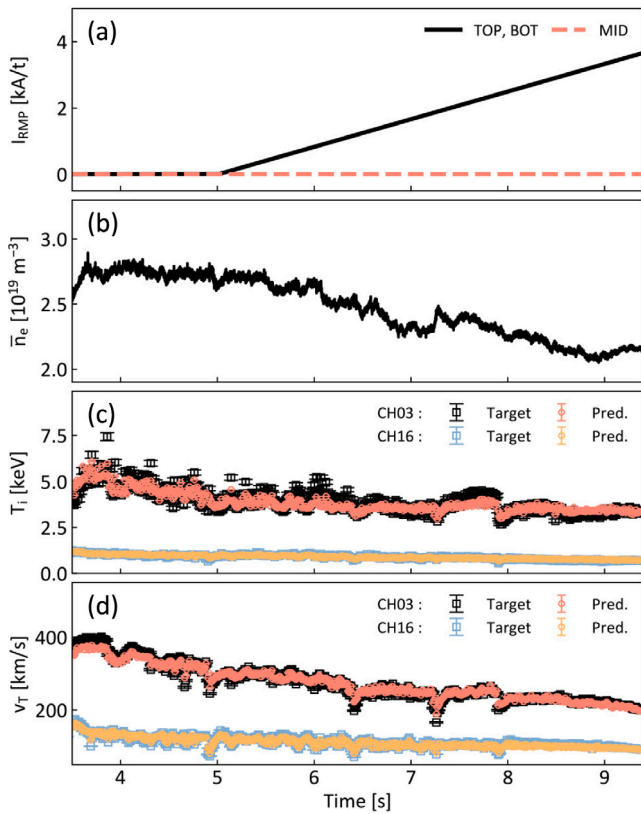


Fig. 5. Comparison of the NN-CES predictions (Pred.) against the conventional analysis results (Target) for KSTAR discharge #28275 from the test set. Time evolution of (a) the applied RMP coil currents for the top and bottom (solid black), and mid (dashed red) coils, (b) the line-averaged electron density, (c) the ion temperature, and (d) the toroidal rotation velocity at a core channel (CH03, $R = 1.853$ m) and an edge channel (CH16, $R = 2.192$ m).

the background level, ensuring that the main spectral features are retained.

To further verify how the background is treated, Fig. 8 compares the target background spectra and the NN-CES predictions at a core channel (CH03) and an edge channel (CH16). In both cases, the predicted background agrees well with the target. Its magnitude increases toward the plasma edge, consistent with the expected contribution from thermal neutrals. This indicates that NN-CES reproduces the background component in a physically consistent manner.

Note that the robust performance shown in this specific case study may differ from the broader statistical R^2 values presented in Fig. 4. The latter reflects an average performance across a diverse test set including various transient phenomena, whereas this case demonstrates the model's high fidelity during relatively stable phases.

5. Discussion and future works

The NN-CES approach presented in this work demonstrates robust performance in inferring ion temperature and toroidal rotation velocity from CES measurements without requiring beam modulation. The model successfully captures the temporal evolution and radial profiles of these plasma parameters, even under complex conditions such as RMP application. The close agreement between NN-CES predictions and results obtained from conventional beam modulation analysis highlights the effectiveness of the proposed neural network approach.

Furthermore, the NN-CES approach may offer an advantage in terms of result stability. By inferring parameters directly from raw spectra in a single end-to-end process, it bypasses the separate background

subtraction and nonlinear fitting steps, where statistical noise and fitting uncertainties can accumulate and propagate. The neural network, acting as a single comprehensive function, can thus produce more stable predictions with lower variance. However, this enhanced stability should not be confused with higher accuracy. Since the model is trained to reproduce the conventional analysis, it will also inherit any potential, underlying errors in those results. A future comparison against systems using a dedicated diagnostic neutral beam (DNB) could be a valuable way to further assess and enhance the model's performance.

The model does have limitations in conditions with challenging signal quality. As the statistical analysis in Fig. 4 indicates, predictive reliability decreases in the edge region. This is attributed to the signal conditions at the plasma boundary, where an increased background signal from thermal neutrals leads to a low active-to-passive signal ratio, resulting in higher uncertainty in the training data. This challenge is not unique to NN-CES but reflects the fundamental constraints of the CES diagnostic itself in high-background regions.

A key advantage of the NN-CES model is its computational efficiency, which was benchmarked against the conventional method using the test set. The conventional analysis, which requires a nonlinear least-squares fit on each background-subtracted spectrum, averaged approximately 2 ms per frame on a CPU (Intel Core i9-14900KF) using Python's `scipy.optimize.curve_fit` function. In contrast, when processing the same test set spectra in a single batch, the average inference time for the NN-CES model on a GPU (NVIDIA GeForce RTX 5070 Ti) was approximately 210 ns per frame. While both methods offer rapid computation on individual spectra, the crucial advantage of NN-CES is the elimination of the operational latency required for beam modulation: the conventional method should wait for a beam-off phase to acquire a background spectrum – a delay of up to 1 second in KSTAR – thereby precluding real-time analysis. By operating directly on raw spectra, the NN-CES method enables continuous, modulation-free measurements.

The present NN-CES model is specific to the C VI line ($\lambda_0 = 529.05$ nm), as the current KSTAR CES system is dedicated to this wavelength. The need to analyze other ion species is becoming increasingly important, particularly with the recent upgrade to a tungsten divertor in KSTAR [16]. Notably, the recent installation of a new spectrometer (Bunkoukeiki Co., Ltd., CLP-300SP) with an adjustable wavelength range paves the way for future work. By acquiring a sufficient training dataset for emission lines with this new instrument, the NN-based approach presented here has the potential to be extended to a broader range of plasma ions.

The ultimate goal is to integrate these capabilities into a comprehensive real-time CES (RT-CES) system. Such a system, capable of providing continuous ion profiles would supply critical inputs for advanced plasma control applications, such as beta control and mode locking detection.

CRediT authorship contribution statement

J.K. Lee: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **W.H. Ko:** Resources, Project administration, Data curation. **H.H. Lee:** Writing – review & editing, Methodology, Investigation, Formal analysis. **B. Kim:** Software, Methodology, Investigation, Formal analysis, Conceptualization. **Y.H. Lee:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **G.W. Shin:** Writing – review & editing, Methodology, Conceptualization. **J. Kim:** Methodology, Formal analysis. **J.M. Lee:** Writing – original draft.

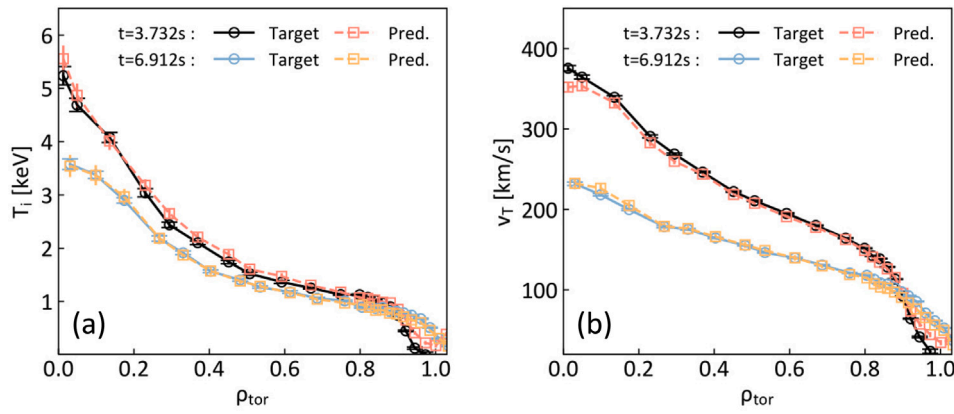


Fig. 6. Radial profiles of (a) ion temperature and (b) toroidal rotation velocity against the normalized toroidal magnetic flux coordinate, ρ_{tor} . Profiles are shown for two time points in #28275: an H-mode phase at $t = 3.732$ s (black solid lines with circles for Target, red dashed lines with squares for Prediction) and a later phase with RMP application at $t = 6.912$ s (blue solid lines with circles for Target, orange dashed lines with squares for Prediction).

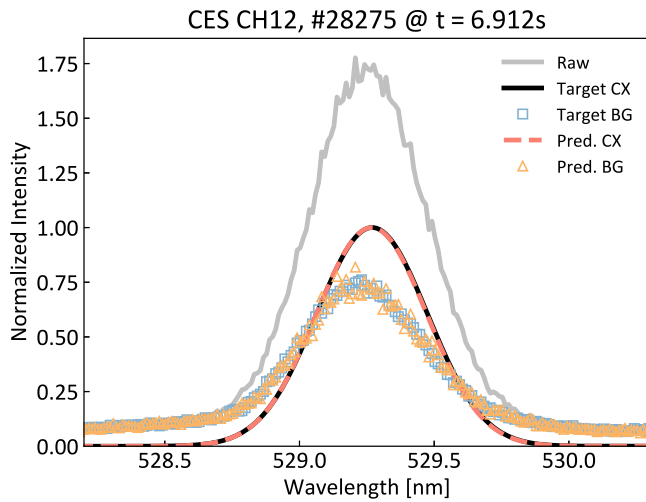


Fig. 7. Comparison of the spectral components for CH12 ($R = 2.147$ m) at $t = 6.912$ s in #28275. The raw spectrum (gray solid line) and target background (Target BG, blue squares) are normalized by the peak of the target active CX spectrum (Target CX, black solid line). The predicted active CX spectrum (Pred. CX, red dashed line) is reconstructed from the model's inferred parameters, and the predicted background (Pred. BG, orange triangles) is subsequently derived by subtracting the predicted CX from the raw signal.

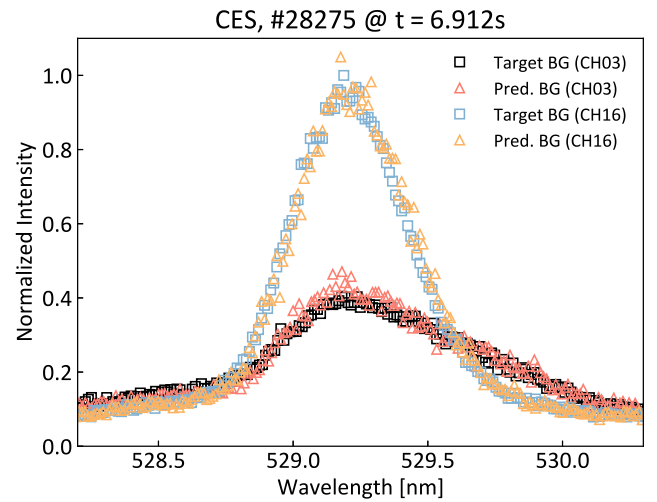


Fig. 8. Comparison of the target background (Target BG) and the predicted background (Pred. BG) spectra for a core channel (CH03, $R = 1.853$ m) and an edge channel (CH16, $R = 2.192$ m) at $t = 6.912$ s in #28275. All spectra are normalized to the peak intensity of the target background spectrum from the edge channel to enable a direct comparison of their relative magnitudes.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jekil Lee reports financial support was provided by Korea Institute of Fusion Energy. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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